

ALEC S. ZABEL-MENA

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EDUCATION

Universtiy of Puerto Rico, at Río Piedras
Doctor of Philosophy of Science in Pure Mathematics

August 2026 – Present

Universtiy of Puerto Rico, at Río Piedras
Master of Science in Pure Mathematics

August 2022 – August 2026

Universtiy of Puerto Rico, at Río Piedras
Bachelor of Science in Pure Mathematics
Minor in Cybersecurity

August 2016 – June 2022

CAREER OBJECTIVE

To become an academic researcher and professor. Research goals include topics in algebraic geometry, matroid theory, and other areas of pure and discrete mathematics.

RESEARCH INTERESTS

- Commutative Algebra
- Algebraic Geometry
- Finite Fields
- Graph theory, and Combinatorics
- Matroid theory
- Category Theory and Type Theory

PUBLICATIONS

1. Zabel-Mena, A.S., 2026, University of Puerto Rico, Río Piedras, *Studies in Absolute Irreducibility over Finite Fields: Rational Points, Singular Zeta-Functions, APN Functions, and Exceptional Polynomials*, master's thesis.
2. Agrinoni, C.A., Janwa, H., Zabel-Mena, A.S. *Absolute Irreducibility of Generalized Trinomials Defined by APN Functions of The Form $g(x) = x^{2^i e} + h(x)$ over $\mathbb{F}_{2^s}$* . Springer Proceedings in Mathematics & Statistics, (Submitted). 2025.

TALKS, SEMINARS, AND COLLOQUIUMS

1. Zabel-Mena, A.S., Janwa, H., Agrinoni, C.A. *Finding Exceptional Trinomials of Given Degree-Gap over $\mathbb{F}_{q^2}$ and Applications to Permutation Polynomials* (Contributed Talk). 41th Seminario Interuniversitario de Investigación en Ciencias Matemáticas (SIDIM), Río Piedras, P.R. 2026.
2. Zabel-Mena, A.S., Janwa, H., Agrinoni, C.A. *Absolute Irreducibility of Generalized Trinomials Defined by APN Functions of The Form $g(x) = x^i + h(x)$ over $\mathbb{F}_{2^s}$* (Contributed Talk). 40th Seminario Interuniversitario de Investigación en Ciencias Matemáticas (SIDIM), Ponce P.R. 2025.
3. Zabel-Mena, A.S., Janwa, H., Agrinoni, C.A. *Absolute Irreducibility of Generalized Trinomials Defined by APN Functions of The Form $g(x) = x^i + h(x)$ over $\mathbb{F}_{2^s}$* (Contributed Talk). 56th Southeastern International Conference on Combinatorics, Graph Theory, & Computing (SEICCGTC), Boca Raton, Florida. 2025.
4. Zabel-Mena, A.S., Janwa, H., Agrinoni, C.A. *An Efficient Implementation and Analysis Of Zeta Functions Of APN Curves and Their Protograph LDPC Codes* (Contributed Talk). 39th Seminario Interuniversitario de Investigación en Ciencias Matemáticas (SIDIM), Humacao P.R. 2024.
5. Zabel-Mena, A.S., Janwa, H., Agrinoni, C.A. *An Efficient Implementation and Analysis Of Zeta Functions Of APN Curves and Their Protograph LDPC Codes* (Poster Session). Forward Research and Innovation Summit, San Juan P.R. 2023.

GRANTS AND AWARDS

Research Assitanship, NASA Grant No. 80NSSC20M0052	August 2025 – December 2025
PR Space Grant Fellowship and Scholarship Program, NASA Grant No. 80NSSC20M0052	August 2024 – May 2025
Research Assitanship, NASA Grant No. 80NSSC20M0052	January 2024 – May 2024
Research Assitanship, NASA Grant No. 80NSSC20M0052	August 2023 – December 2023
PR Space Grant Fellowship and Scholarship Program, NASA Grant No. 80NSSC20M0052	August 2022 – May 2023

TEACHING

Proyecto Tutorías DECEP University of Puerto Rico, Río Piedras Mathematics tutor	2020-2021
Conducted assessments to identify the educational needs of my students and developed individualized learning plans.	
Tasked with providing tutoring services to seven high school students for 10 hours a week in the subjects of Algebra and Pre-Calculus.	
Self-Employed Mathematics tutor	2019-2021
Evaluated students learning styles and provided appropriate techniques for maximizing understanding and minimizing frustration.	
Simplified math concepts while coaching students to think critically when problem solving; eventually introducing them to axiomatic systems within mathematics.	
Provided tutoring one-on-one to five highschool students in the subjects of Algebra, Pre-Calculus, and Geometry for two times a week, at two hours for each session.	

AFFILIATIONS

Consejo General de Estudiantes (URP-RP) Representante ante el Consejo de Estudios Graduados e Investigación	August 2024 – May 2025
Consejo de Estudiantes de Ciencias Naturales (URP-RP) Representante ante el Decanato de Estudios Graduados e Investigación	August 2024 – May 2025
Consejo de Estudiantes de Ciencias Naturales (URP-RP) Representante Graduado del Departamenteo de Matemáticas	August 2023 – May 2024
Asociación Estudiantil de Matemáticas ($A\epsilon M\alpha\tau$) Treasurer	2019 – Present August 2025 – May 2026 August 2022 – May 2024

Asociación de Estudiantes de Ciencias de Computos (AECC)

2019 – Present

Lembrança Negra Capoeira

Corda Crua-Verde

2017 – 2019

Senzala Capoeira (Sou Capoeira)

Corda Amarela-Laranja

2019 – Present

SKILLS

Languages

English

Spanish

Portuguese

Software

L^AT_EX

C programming

Python programming language

SAGE mathematics Software

linux

nix declaritive package manager.